

Notes: Harford, Klasa, and Maxwell (JF, 2014)

Refinancing Risk and Cash Holdings

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	3
3.1	Sample construction	3
3.2	Cash holdings	3
3.3	Refinancing risk and debt maturity	3
3.4	Credit market conditions	4
3.5	Credit lines	4
3.6	Investment and market valuation variables	4
4	Models and empirical specifications	4
4.1	Single-equation debt-maturity trend model	4
4.2	Simultaneous cash and debt maturity model	5
4.3	Cash-flow savings model	5
4.4	Investment model	5
4.5	Market valuation model	5
5	Identification and empirical design	5
5.1	Main identifying logic	5
5.2	Why the 2SLS framework matters	6
5.3	Heterogeneity tests	6
5.4	Limitations	6
6	Tables and figures	7
6.1	Tables I and II: trends in debt maturity and determinants	7
6.2	Tables III and IV: summary statistics and main 2SLS results	7
6.3	Tables VI, VII, and VIII: credit conditions and credit lines	9
6.4	Tables IX, X, and XI: cash savings, investment, and valuation	12
7	Short summary	13
8	Conclusion	14
8.1	Core conclusion	14
8.2	Economic intuition	14
8.3	Takeaway	14

1 Big picture

- **Question.** Do firms hold more cash when they face greater refinancing risk, and can the shortening of corporate debt maturities explain part of the secular rise in cash holdings?
- **Core mechanism.** Shorter maturity debt forces firms to return to capital markets more frequently. If market conditions deteriorate or lenders refuse to roll over debt, firms risk costly refinancing, forced asset sales, inefficient liquidation, or underinvestment. Cash reserves reduce these risks.
- **Main measure of refinancing risk.** The baseline proxy is the fraction of long-term debt due within the next three years, scaled by total long-term debt.
- **Key empirical finding.** Firms with more long-term debt maturing soon hold significantly more cash. In the main 2SLS model, a higher fraction of debt due in the next three years strongly increases the log of cash-to-assets.
- **Trend fact.** Long-term debt maturities shortened substantially from 1980 to 2008, while corporate cash holdings rose. The paper estimates that shorter debt maturity explains roughly one-third of the rise in cash holdings documented in the corporate cash literature.
- **Joint financial policy.** Cash holdings and debt maturity are jointly determined. Firms that use shorter maturity debt hold cash as a hedge against refinancing risk, and cash may also make shorter maturity debt more feasible, though the latter effect is economically small.
- **When the effect is strongest.** The cash–debt–maturity relation is stronger when firms have more debt, when credit conditions are weak or worsening, and when credit lines are less reliable substitutes for cash.
- **Real effects.** Cash is especially valuable for firms with high refinancing risk because it mitigates underinvestment. Cash-rich, high-refinancing-risk firms invest more than otherwise similar firms, especially in tight credit markets.
- **Valuation effect.** The market value of cash is higher for firms facing refinancing risk. Investors appear to value cash more when it protects the firm against rollover problems.
- **Economic takeaway.** Debt maturity and cash policy are interdependent. The rise in corporate cash cannot be understood without accounting for the simultaneous shortening of debt maturity and the associated refinancing risk.

2 Incremental contribution of the paper

- **Contribution relative to cash-holdings papers.** Earlier work explains high cash holdings using transaction motives, precautionary motives, agency problems, taxes, and growth opportunities. This paper adds refinancing risk from debt maturity structure as a quantitatively important determinant of cash holdings.
- **Contribution relative to debt-maturity papers.** Debt maturity research studies why firms choose short-term or long-term debt. This paper shows that debt maturity choices cannot be analyzed in isolation because they alter the firm’s optimal liquidity policy.
- **Contribution relative to studies of the rise in cash.** Bates, Kahle, and Stulz document the secular rise in cash. Harford, Klasa, and Maxwell show that a large portion of this rise is linked to the shortening of long-term debt maturities.
- **Contribution relative to refinancing-risk theory.** Theories by Diamond, Sharpe, Froot–Scharfstein–Stein, and others emphasize rollover risk and underinvestment. This paper provides firm-level empirical evidence that firms respond to these risks by holding cash.

- **Methodological contribution.** The paper explicitly treats cash holdings and debt maturity as jointly determined, using a simultaneous-equations framework rather than a single-equation cash regression.
- **New channel through credit lines.** The paper asks whether unused credit lines substitute for cash as refinancing-risk insurance. The answer is nuanced: credit lines reduce the need for cash only for financially unconstrained firms; constrained firms still rely heavily on cash.
- **New valuation test.** The paper connects refinancing risk to the marginal value of cash, showing that shareholders value cash more when firms are exposed to rollover risk.

3 Data and measurement

3.1 Sample construction

- The main sample uses Compustat industrial firms from 1980 to 2008.
- Financial firms and utilities are excluded.
- Firms must be U.S.-incorporated, have nonzero sales and assets, and have positive long-term debt for most debt-maturity tests.
- Table I uses 103,806 firm-year observations for broad debt and cash trends.
- The main simultaneous-equations regressions use 80,035 firm-year observations with the variables required for the cash and maturity equations.

Interpretation: The focus is on nonfinancial firms because the maturity structure of financial firms is fundamentally different and because short-term liabilities play a special role in financial intermediation.

3.2 Cash holdings

The main dependent variable in the cash regressions is

$$\log \left(\frac{Cash + ShortTermInvestments}{BookAssets} \right). \quad (1)$$

- Cash and short-term investments are measured using Compustat cash variables.
- Scaling by book assets makes cash holdings comparable across firms of different sizes.
- The logarithm is used because the cash-to-assets ratio is highly skewed.

Interpretation: A positive coefficient in the cash equation means that a variable increases the proportional level of cash holdings, not merely the level of cash dollars.

3.3 Refinancing risk and debt maturity

The main refinancing-risk proxy is

$$RefRisk_{i,t} = \frac{LongTermDebtDueWithin3Years_{i,t}}{TotalLongTermDebt_{i,t}}. \quad (2)$$

- Total long-term debt includes debt maturing in more than one year plus the current portion of long-term debt.
- The numerator includes long-term debt due in the next three years.
- The paper also checks one-year and five-year maturity windows.

Interpretation: A high value of this variable means that a large portion of the firm’s long-term debt must be refinanced soon. The paper uses long-term debt rather than total debt because short-term debt is often used for working-capital matching and is partly captured through net working capital controls.

3.4 Credit market conditions

The paper measures broad credit-market conditions using the spread between commercial and industrial loan rates and the federal funds rate.

- Strong credit markets are years with spreads below the median.
- Less strong credit markets are years with spreads at or above the median.
- Weakening credit markets are years in which the spread is high but not in the five highest-spread years.
- Tight credit market years are the five years with the highest spreads.

Interpretation: Refinancing risk should matter most when credit conditions are poor, because debt rollover is more costly and uncertain.

3.5 Credit lines

The paper collects data on credit lines from 1996 to 2008 and constructs variables for:

- whether a firm uses at least some of its credit line;
- whether a firm uses only part of its credit line;
- the unused amount of credit-line capacity.

Interpretation: Credit lines are potential liquidity substitutes. The key question is whether they fully replace cash as insurance against refinancing risk.

3.6 Investment and market valuation variables

The paper measures investment as capital expenditures plus acquisitions, R&D, and advertising, scaled by book assets. It also estimates market value regressions following the valuation framework of Faulkender and Wang.

Interpretation: These tests move beyond the existence of cash hoarding and ask whether cash actually helps firms avoid underinvestment and whether shareholders value cash more when refinancing risk is high.

4 Models and empirical specifications

4.1 Single-equation debt-maturity trend model

The paper first estimates whether debt maturities have shortened after controlling for firm characteristics:

$$RefRisk_{i,t} = \alpha + \beta Year_t + \gamma' X_{i,t} + IndustryFE + \varepsilon_{i,t}. \quad (3)$$

- $RefRisk_{i,t}$ is the fraction of long-term debt due within the next three years.
- $X_{i,t}$ includes leverage, size, market-to-book, term structure, future earnings, weighted asset maturity, cash flow risk, net debt issuance, IPO status, and sometimes cash holdings.

Interpretation: A positive coefficient on the year variable means that, conditional on firm characteristics, debt maturity is shortening over time.

4.2 Simultaneous cash and debt maturity model

The main empirical design treats cash and debt maturity as jointly determined:

$$\log(\text{CashAssets}_{i,t}) = \alpha_1 + \beta_1 \text{RefRisk}_{i,t} + \gamma'_1 X_{i,t} + u_{i,t}, \quad (4)$$

$$\text{RefRisk}_{i,t} = \alpha_2 + \beta_2 \log(\text{CashAssets}_{i,t}) + \gamma'_2 Z_{i,t} + v_{i,t}. \quad (5)$$

The paper estimates this system using 2SLS.

Interpretation: This is essential because firms that hold more cash may also be better able to issue or accept shorter maturity debt. The system isolates the effect of maturity on cash after accounting for the reverse channel.

4.3 Cash-flow savings model

To test whether firms with refinancing risk save more cash from current cash flows, the paper estimates:

$$\Delta \text{Cash}_{i,t} = \alpha + \beta \text{CashFlow}_{i,t} + \delta \text{HighRefRisk}_{i,t} + \phi(\text{CashFlow}_{i,t} \times \text{HighRefRisk}_{i,t}) + \Gamma' X_{i,t} + \varepsilon_{i,t}. \quad (6)$$

Interpretation: A positive interaction coefficient means firms facing high refinancing risk save a larger fraction of cash flow rather than spending or distributing it.

4.4 Investment model

To test underinvestment, the paper estimates:

$$\text{Investment}_{i,t} = \alpha + \beta \text{Cash}_{i,t-1} + \delta \text{HighRefRisk}_{i,t} + \phi(\text{Cash}_{i,t-1} \times \text{HighRefRisk}_{i,t}) + \Gamma' X_{i,t} + \varepsilon_{i,t}. \quad (7)$$

Interpretation: If cash mitigates underinvestment, the interaction between lagged cash and high refinancing risk should be positive.

4.5 Market valuation model

Following Faulkender and Wang, the paper estimates a change-in-value specification:

$$\Delta \text{FirmValue}_{i,t} = \alpha + \beta \Delta \text{Cash}_{i,t} + \delta \text{HighRefRisk}_{i,t} + \phi(\text{HighRefRisk}_{i,t} \times \Delta \text{Cash}_{i,t}) + \Gamma' X_{i,t} + \varepsilon_{i,t}. \quad (8)$$

Interpretation: The coefficient on the interaction measures whether an additional dollar of cash is worth more when firms face high refinancing risk.

5 Identification and empirical design

5.1 Main identifying logic

- The key empirical variation is cross-sectional and time-series variation in debt maturity.
- The paper controls for a large set of firm characteristics that jointly affect cash and debt maturity.
- Industry and year controls absorb broad industry characteristics and macroeconomic trends.
- The simultaneous-equations approach addresses the reverse relation from cash holdings to debt maturity.

Interpretation: The paper does not use a single quasi-random shock. Instead, it builds a consistent body of evidence across several tests: trends, 2SLS cash equations, credit-condition heterogeneity, credit lines, cash saving, investment, and valuation.

5.2 Why the 2SLS framework matters

- Cash holdings and debt maturity are financial policy choices that may be made jointly.
- If researchers regress cash on debt maturity without accounting for simultaneity, the coefficient can combine the effect of maturity on cash and the effect of cash on maturity.
- The paper reports Hausman tests indicating that joint determination is empirically relevant.
- The main conclusion remains: debt maturing soon increases cash holdings.

Interpretation: Treating maturity as exogenous would understate or confound the role of refinancing risk. The simultaneous system is a core contribution of the paper.

5.3 Heterogeneity tests

The refinancing-risk interpretation predicts stronger effects when refinancing is more costly:

- firms with more debt should respond more strongly;
- effects should be stronger in weak credit markets;
- credit lines should attenuate the need for cash mainly for unconstrained firms;
- cash should matter more for investment and valuation among high-refinancing-risk firms.

Interpretation: These heterogeneity tests are important because they test the mechanism, not just a statistical association.

5.4 Limitations

- Debt maturity and cash policies are not randomized.
- Credit-line data are available only for a shorter subperiod.
- The maturity measure captures contractual debt coming due, but cannot perfectly measure rollover difficulty.
- The 2SLS framework helps with simultaneity but relies on model specification and controls.

Interpretation: The paper is strongest as a multi-test corporate-finance design. Its credibility comes from consistency across several outcomes and subsamples.

6 Tables and figures

6.1 Tables I and II: trends in debt maturity and determinants

Read:

- Table I documents the shortening of debt maturity: the fraction of long-term debt due within three years rises from 0.383 in 1980–1984 to 0.427 in 2005–2008.
- Cash holdings rise over the same period from 0.110 to 0.177 of book assets.
- Table II shows that the shortening of debt maturity remains after controlling for leverage, size, asset maturity, cash flow risk, term structure, and other firm characteristics.

Economic message: The rise in corporate cash coincides with a marked shortening of debt maturity, suggesting refinancing risk as a missing component of the cash-holdings puzzle.

Table I
The Changing Nature of Debt and Cash Holdings in the United States

This table examines the changing nature of debt and cash holdings characteristics of U.S. incorporated firms with nonzero sales and total assets and with long-term debt > 0 over the 1980 to 2008 period. Utilities and financials are excluded. Our final sample includes 103,806 firm-years. Long-term debt is defined as debt maturing in more than one year and the current portion of long-term debt (Compustat variables DLT + DD1). To express the time trends in debt characteristics over time, we split the sample into six time periods and compute yearly means and then calculate the average over the years for each time period. The maturity structure of firms' public and private bonds is calculated with data from the FISD database. The maturity structure of firms' bank debt is calculated with data from the Dealscan database. We merge both databases with Compustat. To calculate the maturity of bonds and bank loans, we collect data at the issue level on the amount of bonds and loans issued each year and then create a value-weighted average maturity of debt for newly issued bonds and bank debt.

	1980 to 1984	1985 to 1989	1990 to 1994	1995 to 1999	2000 to 2004	2005 to 2008
Debt Characteristics						
Proportion of Compustat firms with long-term debt	0.900	0.876	0.844	0.822	0.788	0.763
Leverage ratio	0.229	0.245	0.235	0.243	0.241	0.243
Fraction of long-term debt due within one year	0.149	0.179	0.209	0.208	0.225	0.191
Fraction of long-term debt due within three years	0.383	0.425	0.488	0.470	0.482	0.427
Fraction of long-term debt due within five years	0.535	0.567	0.631	0.620	0.626	0.626
Debt tied to prime/long-term debt	0.168	0.204	0.208	0.232	0.226	0.258
Debentures/long-term debt	0.093	0.096	0.062	0.042	0.036	0.031
Average bond maturity	16.6	13.2	13.5	10.4	11.3	
Average bank loan maturity	5.0	4.1	4.3	3.1	3.8	
Average bond & loan weighted maturity	10.9	6.8	6.9	6.3	5.6	
Cash						
Cash holdings and short-term investments/book assets	0.110	0.124	0.126	0.149	0.176	0.177

Table I: Debt maturity and cash trends

Table II
The Change in Debt Maturity after Controlling for Determinants of Maturity

Data are for Compustat industrial firms over the 1980 to 2008 period. The dependent variable is long-term debt due over the next three years/total long-term debt. Observation year is the fiscal year when an observation takes place. Term structure is the difference between the yield on a government 10-year and six-month bond. Future-year abnormal earnings is the difference between earnings per share in year $t + 1$ (excluding extraordinary items and discontinued operations and adjusted for any changes in shares outstanding) minus earnings per share in year t , divided by the year t share price. Weighted average asset maturity is defined as the book value-weighted maturity of long-term assets and current assets, where the maturity of long-term assets is computed as gross property, plant, and equipment divided by depreciation expense and the maturity of current assets is computed as current assets divided by the cost of goods sold. Industry cash flow risk is calculated as follows. For each firm-year, we compute the standard deviation of cash flow to assets for the previous 10 years, requiring at least three observations. We then average the firm cash flow standard deviations each year across each two-digit SIC industry. Net debt issuance is annual long-term debt issuance minus long-term debt reduction. Industry effects are controlled for by including dummies for Fama-French (1997) 48 industry groups. Significance levels for whether coefficient estimates are different from zero are in parentheses. The standard errors of the coefficients are adjusted for clustering of observations at the firm level.

Model	1	2
Intercept	-15.801 (<0.001)	-15.900 (<0.001)
Observation year	0.008 (<0.001)	0.008 (<0.001)
Total debt/book assets	-0.221 (<0.001)	-0.240 (<0.001)
Natural logarithm of real book value of assets	-0.058 (<0.001)	-0.059 (<0.001)
Market-to-book assets	0.000 (0.748)	0.002 (0.074)
Term structure	0.004 (<0.001)	0.004 (<0.001)
Future year abnormal earnings	0.030 (<0.001)	0.031 (<0.001)
Weighted average asset maturity	-0.003 (<0.001)	-0.003 (<0.001)
Industry cash flow risk	0.043 (0.212)	0.051 (0.142)
Net debt issuance/book assets	-0.328 (<0.001)	-0.322 (<0.001)
Firm had its IPO during the prior five years dummy	0.003 (0.418)	0.004 (0.357)
Natural logarithm of cash and short-term investments/book assets		-0.011 (0.000)
Industry fixed effects	Yes	Yes
R^2 -adjusted	0.198	0.200
N	80,035	80,035

Table II: Determinants of debt maturity

6.2 Tables III and IV: summary statistics and main 2SLS results

Read:

- Table III reports that average cash holdings are 0.124 and average debt due within three years is 0.400 in the regression sample.
- The fraction of firms with all long-term debt due within three years rises from 0.028 in 1980 to 0.124 in 2008.
- Table IV shows that shorter maturity debt significantly increases cash holdings across several maturity definitions.

- The main coefficient implies that a 1% increase in the fraction of long-term debt due within three years raises cash holdings by about 2.4%.

Economic message: The main 2SLS evidence shows that the maturity structure of debt is an economically important determinant of corporate cash policy.

Table III
Univariate Characteristics of Sample Used for Multivariate Tests

Panels A, B, and C report descriptive statistics using the sample of 80,035 firm-years for which it is possible to calculate the dependent and independent variables used in the regression models in Tables IV and VI. Panel D reports for each year from 1980 to 2008 the four-quarter moving average of the spread of the commercial and industrial loan rates over the federal funds rate.

Sample Period	Mean	25 th Pct.	Median	75 th Pct.	Fraction of Firms with All Long-Term Debt Due within Three Years
Panel A: 1980 to 2008					
Cash holdings and short-term investments/book assets	0.124	0.019	0.058	0.160	–
Fraction of long-term debt due within three years	0.400	0.110	0.314	0.653	0.089
Panel B: 1980					
Cash holdings and short-term investments/book assets	0.085	0.022	0.049	0.106	–
Fraction of long-term debt due within three years	0.338	0.155	0.274	0.456	0.028
Panel C: 2008					
Cash holdings and short-term investments/book assets	0.139	0.024	0.074	0.183	–
Fraction of long-term debt due within three years	0.394	0.040	0.293	0.678	0.124
Panel D: Four-Quarter Moving Average of the Spread of Commercial and Industrial Loan Rates over the Federal Funds Rate					
Year	Spread	Year	Spread	Year	Spread
1980	2.01	1990	1.53	2000	1.80
1981	2.73	1991	1.75	2001	1.71
1982	1.87	1992	1.63	2002	1.79
1983	1.23	1993	1.57	2003	2.07
1984	1.24	1994	1.37	2004	1.95
1985	0.83	1995	1.37	2005	2.08
1986	0.97	1996	1.23	2006	1.73
1987	1.42	1997	1.31	2007	1.59
1988	1.54	1998	1.38	2008	2.12
1989	1.73	1999	1.64		

Table III: Summary statistics

Table IV
The Effect of Debt Maturity on Cash Holdings

Data are for Compustat industrial firms over the 1980 to 2008 period. The table reports the second-stage results for the structural equation that explains cash holdings estimated using the 2SLS methodology. The second-stage results for the structural equation that explains debt maturity are reported in the Internet Appendix. The second-stage structural equation that explains debt maturity has long-term debt due over the next three years/total long-term debt as the dependent variable, and the independent variables for this model are the predicted value of the natural logarithm of the sum of cash and short-term investments/book assets, total debt/book assets, the natural logarithm of real book assets, market-to-book assets, the difference between the yield on a government 10-year and six-month bond, future-year abnormal earnings, weighted average maturity of a firm's assets, industry cash flow risk, net debt issuance/book assets, whether a firm had its IPO during the prior five years, the average commercial and industrial loan rate spread (spread above the federal funds rate) over a firm's fiscal year, Fama-French (1997) 48 industry dummy variables, and year dummy variables. Industry cash flow risk is calculated as follows. For each firm-year, we compute the standard deviation of cash flow to assets for the previous 10 years, requiring at least three observations. We then average the firm cash flow standard deviations each year across each two-digit SIC industry. Net debt issuance is annual long-term debt issuance minus long-term debt reduction. Industry fixed effects are dummies for Fama-French (1997) 48 industry groups. Significance levels for whether coefficient estimates are different from zero are in parentheses. The standard errors of the coefficients are adjusted for clustering of observations at the firm level.

Model	Sample Period 1980 to 2008				Sample Period 1980 to 2006
	1	2	3	4	5
Intercept	–6.789 (<0.001)	–3.910 (<0.001)	–8.492 (<0.001)	–6.800 (<0.001)	–6.589 (<0.001)
Long-term debt due in next three years/total long-term debt	5.954 (<0.001)				5.887 (<0.001)
Long-term debt due in next year/total long-term debt		5.122 (<0.001)			
Long-term debt due in next five years/total long-term debt			6.811 (<0.001)		
Debt due in next three years/total debt				6.614 (<0.001)	
Net working capital/book assets	–1.867 (<0.001)	–1.506 (<0.001)	–2.190 (<0.001)	–4.017 (<0.001)	–1.844 (<0.001)
Natural logarithm of real book value of assets	0.285 (<0.001)	0.112 (<0.001)	0.291 (<0.001)	0.252 (<0.001)	0.278 (<0.001)
Market-to-book assets	0.122 (<0.001)	0.092 (<0.001)	0.155 (<0.001)	0.085 (<0.001)	0.121 (<0.001)
Research and development expenses/sales	0.593 (<0.001)	0.579 (<0.001)	0.588 (<0.001)	0.659 (<0.001)	0.589 (<0.001)
Capital expenditures/book assets	–1.750 (<0.001)	–1.272 (<0.001)	–2.331 (<0.001)	–2.175 (<0.001)	–1.756 (<0.001)
Dividend paying dummy	–0.099 (0.018)	–0.131 (<0.001)	–0.142 (0.004)	–0.127 (0.004)	–0.081 (0.061)
Operating income/book assets	0.663 (<0.001)	0.951 (<0.001)	0.350 (<0.001)	1.321 (<0.001)	0.645 (<0.001)
Total debt/book assets	–0.701 (<0.001)	–1.084 (<0.001)	–1.000 (<0.001)	–0.564 (<0.001)	–0.785 (<0.001)

Table IV: Main cash equation

Table IV—Continued					
Model	Sample Period 1980 to 2008				Sample Period 1980 to 2006
	1	2	3	4	5
Industry cash flow risk	0.267 (0.309)	0.112 (0.590)	0.705 (0.031)	−0.075 (0.808)	0.506 (0.102)
Acquisition expense/book assets	−2.092 (<0.001)	−1.838 (<0.001)	−3.182 (<0.001)	−2.215 (<0.001)	−2.147 (<0.001)
Commercial and industrial loan rate spread	0.021 (0.636)	0.007 (0.820)	−0.022 (0.673)	0.014 (0.775)	0.019 (0.663)
Net debt issuance/book assets	3.084 (<0.001)	2.356 (<0.001)	3.382 (<0.001)	3.600 (<0.001)	3.122 (<0.001)
Firm had its IPO during the prior five years dummy	0.034 (0.270)	−0.032 (0.163)	0.048 (0.199)	0.027 (0.410)	0.035 (0.264)
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
N	80,035	80,035	80,035	80,035	76,398

Table IV continued

Table V Predicted Debt and the Effect of Debt Maturity on Cash Holdings		
Data are for Compustat industrial firms over the 1980 to 2008 period. The table reports the second-stage results for the structural equation that explains cash holdings estimated using the 2SLS methodology. The variables and the debt maturity model that has long-term debt due over the next three years/total long-term debt as the dependent variable are defined in Table IV. A firm's predicted book leverage is the fitted value from a regression of book leverage on lagged market-to-book assets; lagged property, plant, and equipment/book assets; lagged research and development expenses/sales; a dummy variable for whether a firm has no research and development expenses; lagged selling expenses/sales; the lagged value of the natural logarithm of sales; Fama-French (1997) 48 industry dummy variables, and year dummy variables. Significance levels for whether coefficient estimates are different from zero are in parentheses. The standard errors of the coefficients are adjusted for clustering of observations at the firm level.		
Model	Predicted Debt Level > Annual Sample Median Value 1	Predicted Debt Level ≤ Annual Sample Median Value 2
Intercept	−7.718 (<0.001)	−2.281 (<0.001)
Long-term debt due in next three years/total long-term debt	6.397 (<0.001)	1.620 (0.019)
Net working capital/book assets	−1.216 (<0.001)	−2.681 (<0.001)
Natural logarithm of real book value of assets	0.304 (<0.001)	0.040 (0.030)
Market-to-book assets	0.103 (<0.001)	0.116 (<0.001)
Research and development expenses/sales	0.572 (<0.001)	0.577 (<0.001)
Capital expenditures/book assets	−1.326 (<0.001)	−2.775 (<0.001)
Dividend paying dummy	−0.032 (0.596)	−0.223 (<0.001)
Operating income/book assets	0.300 (0.012)	0.557 (<0.001)
Total debt/book assets	−0.308 (0.139)	−2.099 (<0.001)
Industry cash flow risk	1.131 (0.003)	−0.205 (0.380)
Acquisition expense/book assets	−2.041 (<0.001)	−2.569 (<0.001)
Commercial and industrial loan rate spread	0.005 (0.946)	−0.025 (0.533)
Net debt issuance/book assets	2.898 (<0.001)	2.093 (<0.001)
Firm had its IPO during the prior five years dummy	−0.032 (0.471)	−0.033 (0.231)
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
N	37,155	37,168

Table V: Predicted debt maturity

6.3 Tables VI, VII, and VIII: credit conditions and credit lines

Read:

- Table VI shows that the debt-maturity effect is much stronger when credit markets are less strong or weakening.
- In tight credit markets, the coefficient on debt due within three years is not statistically significant, consistent with sample composition and severe credit rationing.
- Table VII shows that the positive relation between debt maturity and cash remains in the credit-line subsample.
- Table VIII shows that credit lines do not fully substitute for cash, especially among financially constrained firms.

Economic message: Firms appear to use cash as protection against rollover risk precisely when credit market conditions or credit-line access make refinancing more uncertain.

Table VI
Credit Market Conditions and the Effect of Debt Maturity
on Cash Holdings

Data are for Compustat industrial firms over the 1980 to 2008 period. The table reports the second-stage results for the structural equation that explains cash holdings estimated using the 2SLS methodology. The variables and the debt maturity model that has long-term debt due over the next three years/total long-term debt as the dependent variable are defined in Table IV. The first model reports the results for the years when the commercial and industrial loan rate spread was less than the median value of 1.63 for the 29 years from 1980 to 2008. The second model reports the results for the years when the commercial and industrial loan rate spread was greater than or equal to the median value of 1.63 for the 29 years from 1980 to 2008. The third model reports the results for the years from 1980 to 2008 when the commercial and industrial loan rate spread is greater than or equal to the median value of 1.63 for the 29 years from 1980 to 2008, but excluding the five years with the highest spread values. The fourth model reports the results for the five years from 1980 to 2008 with the highest commercial and industrial loan rate spread values. Significance levels for whether coefficient estimates are different from zero are in parentheses. The standard errors of the coefficients are adjusted for clustering of observations at the firm level.

Model	Credit Market Is		Less Strong Credit Market Is	
	Strong 1	Less Strong 2	Weakening 3	Tight 4
Intercept	-1.605 (<0.001)	-6.032 (<0.001)	-9.728 (<0.001)	-2.246 (<0.001)
Long-term debt due in next three years/total long-term debt	0.669 (0.041)	4.389 (<0.001)	6.236 (<0.001)	0.515 (0.132)
Net working capital/book assets	-2.010 (<0.001)	-1.837 (<0.001)	-1.720 (<0.001)	-2.345 (<0.001)
Natural logarithm of real book value of assets	0.038 (0.032)	0.117 (<0.001)	0.201 (<0.001)	-0.036 (0.007)
Market-to-book assets	0.106 (<0.001)	0.118 (<0.001)	0.119 (<0.001)	0.129 (<0.001)
Research and development expenses/sales	0.514 (<0.001)	0.559 (<0.001)	0.606 (<0.001)	0.422 (<0.001)
Capital expenditures/book assets	-1.891 (<0.001)	-1.343 (<0.001)	-0.898 (0.007)	-2.302 (<0.001)
Dividend paying dummy	-0.165 (<0.001)	0.106 (0.036)	0.194 (0.008)	-0.097 (0.016)
Operating income/book assets	0.507 (<0.001)	0.819 (<0.001)	0.893 (<0.001)	0.618 (<0.001)
Total debt/book assets	-2.071 (<0.001)	-0.771 (<0.001)	-0.343 (0.216)	-1.622 (<0.001)
Industry cash flow risk	-0.272 (0.238)	-1.041 (<0.001)	-1.829 (<0.001)	0.197 (0.342)
Acquisition expense/book assets	-1.842 (<0.001)	-2.333 (<0.001)	-2.097 (<0.001)	-2.541 (<0.001)
Commercial and industrial loan rate spread	-0.384 (<0.001)	0.619 (<0.001)	1.967 (<0.001)	0.076 (0.016)
Net debt issuance/book assets	1.392 (<0.001)	2.611 (<0.001)	3.285 (<0.001)	1.152 (<0.001)

(Continued)

Table VI: Credit-market conditions

Table VI—Continued

Model	Credit Market Is		Less Strong Credit Market Is	
	Strong 1	Less Strong 2	Weakening 3	Tight 4
Firm had its IPO during the prior five years dummy	0.007 (0.062)	0.010 (0.763)	-0.006 (0.898)	0.041 (0.252)
Year fixed effects	No	No	No	No
Industry fixed effects	Yes	Yes	Yes	Yes
N	41,586	38,449	27,229	11,220

Table VI continued

Table VII
Credit Lines and the Effect of Debt Maturity on Cash Holdings

Data are for Compustat industrial firms over the 1996 to 2008 period for which we collect data on credit lines. The table reports the second-stage results for the structural equation that explains cash holdings estimated using the 2SLS methodology. The variables and the debt maturity model that has long-term debt due over the next three years/total long-term debt as the dependent variable are defined in Table IV. In the Internet Appendix, we report how we create the credit line variables. The table reports regression results estimated using the 2SLS methodology. Significance levels for whether coefficient estimates are different from zero are in parentheses. The standard errors of the coefficients are adjusted for clustering of observations at the firm level.

Model	All Firms		Firms with a Credit Line	
	1	2	3	4
Intercept	-5.145 (<0.001)	-5.001 (<0.001)	-5.161 (<0.001)	-4.622 (<0.001)
Long-term debt due in next three years/total long-term debt	3.824 (<0.001)	4.099 (<0.001)	4.000 (<0.001)	3.900 (<0.001)
Net working capital/book assets	-2.351 (<0.001)	-2.221 (<0.001)	-2.324 (<0.001)	-2.269 (<0.001)
Natural logarithm of real book value of assets	0.241 (<0.001)	0.267 (<0.001)	0.260 (<0.001)	0.249 (<0.001)
Market-to-book assets	0.133 (<0.001)	0.121 (<0.001)	0.145 (<0.001)	0.131 (<0.001)
Research and development expenses/sales	0.519 (<0.001)	0.466 (<0.001)	0.621 (<0.001)	0.641 (<0.001)
Capital expenditures/book assets	-1.703 (<0.001)	-1.556 (<0.001)	-1.689 (<0.001)	-1.549 (<0.001)
Dividend paying dummy	-0.239 (<0.001)	-0.230 (<0.001)	-0.223 (<0.001)	-0.239 (<0.001)
Operating income/book assets	0.424 (<0.001)	0.514 (<0.001)	0.292 (0.019)	0.186 (0.125)
Total debt/book assets	-0.667 (0.001)	-0.534 (0.009)	-0.710 (0.002)	-0.491 (0.026)
Industry cash flow risk	0.178 (0.478)	0.147 (0.568)	0.062 (0.808)	0.104 (0.670)
Acquisition expense/book assets	-2.400 (<0.001)	-2.271 (<0.001)	-2.333 (<0.001)	-2.146 (<0.001)
Commercial and industrial loan rate spread	-0.117 (0.323)	-0.092 (0.456)	-0.197 (0.125)	-0.156 (0.213)
Net debt issuance/book assets	2.253 (<0.001)	2.299 (<0.001)	2.333 (<0.001)	2.214 (<0.001)
Firm had its IPO during the prior five years dummy	-0.039 (0.290)	-0.040 (0.299)	-0.044 (0.273)	-0.032 (0.410)
Firm has a credit line dummy		-0.596 (<0.001)		
Firm is using some of its credit line dummy				-0.712 (<0.001)
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
N	26,714	26,714	23,502	23,502

Table VII: Credit lines and cash holdings

Table VIII
The Effect of Credit Line Usage on Firms' Propensities to Mitigate Refinancing Risk with Cash Holdings

Data are for Compustat industrial firms over the 1996 to 2008 period for which we collect data on credit lines. The table reports the second-stage results for the structural equation that explains cash holdings estimated using the 2SLS methodology. The variables and the debt maturity model that has long-term debt due over the next three years/total long-term debt as the dependent variable are defined in Table IV. In the Internet Appendix, we report how we create the credit line variables. Significance levels for whether coefficient estimates are different from zero are in parentheses. The standard errors of the coefficients are adjusted for clustering of observations at the firm level.

Panel A: Financially Constrained and Unconstrained Firms Analyzed Together			
Model	1	2	3
Firm is using at least some of its credit line	Yes	No	-
Intercept	-5.171 (<0.001)	-6.215 (<0.001)	-4.698 (<0.001)
Long-term debt due in next three years/total long-term debt	3.835 (<0.001)	4.726 (0.002)	4.059 (<0.001)
Dummy for if a firm is using its credit line			-0.631 (<0.001)
Long-term debt due in next three years/total long-term debt × Dummy for if a firm is using its credit line			-0.201 (0.334)
Net working capital/book assets	-2.320 (<0.001)	-1.921 (<0.001)	-2.272 (<0.001)
Natural logarithm of real book value of assets	0.269 (<0.001)	0.223 (0.023)	0.249 (<0.001)
Market-to-book assets	0.166 (<0.001)	0.051 (0.048)	0.130 (<0.001)
Research and development expenses/sales	0.606 (<0.001)	0.697 (<0.001)	0.641 (<0.001)
Capital expenditures/book assets	-1.476 (<0.001)	-1.685 (0.025)	-1.551 (<0.001)
Dividend paying dummy	-0.211 (<0.001)	-0.421 (<0.001)	-0.240 (<0.001)
Operating income/book assets	0.118 (0.364)	0.566 (0.047)	0.181 (0.135)
Total debt/book assets	-0.660 (0.002)	0.836 (0.324)	-0.484 (0.027)
Industry cash flow risk	0.020 (0.944)	0.308 (0.457)	0.106 (0.665)
Acquisition expense/book assets	-1.841 (<0.001)	-3.573 (<0.001)	-2.156 (<0.001)
Commercial and industrial loan rate spread	-0.229 (0.124)	0.259 (0.353)	-0.152 (0.224)
Net debt issuance/book assets	2.020 (<0.001)	3.032 (<0.001)	2.213 (<0.001)
Firm had its IPO during the prior five years dummy	-0.032 (0.469)	-0.011 (0.887)	-0.034 (0.392)
Year fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes

Table VIII, Panel A: Credit-line usage

Table VIII—Continued

Panel B: Financially Constrained and Unconstrained Firms Analyzed Separately						
Model	1	2	3	4	5	6
Firm has a bond rating	No	No	No	Yes	Yes	Yes
Firm is using at least some of its credit line	Yes	No	–	Yes	No	–
Intercept	–4.963 (<0.001)	–4.642 (0.003)	–4.251 (<0.001)	–1.504 (0.032)	0.135 (0.890)	–0.999 (0.117)
Long-term debt due in next three years/total long-term debt	3.370 (0.002)	3.138 (0.007)	3.228 (<0.001)	3.484 (<0.001)	–1.187 (0.534)	3.758 (<0.001)
Dummy for if a firm is using its credit line			–0.831 (<0.001)			–0.451 (0.004)
Long-term debt due in next three years/total long-term debt × Dummy for if a firm is using its credit line			0.216 (0.442)			–0.525 (0.491)
Net working capital/book assets	–2.367 (<0.001)	–2.154 (<0.001)	–2.346 (<0.001)	–2.040 (<0.001)	–2.260 (<0.001)	–1.930 (<0.001)
Natural logarithm of real book value of assets	0.300 (0.002)	0.142 (0.062)	0.216 (<0.001)	–0.114 (0.002)	–0.180 (<0.001)	–0.128 (<0.001)
Market-to-book assets	0.153 (<0.001)	0.055 (0.013)	0.121 (<0.001)	0.303 (<0.001)	0.164 (<0.001)	0.272 (<0.001)
Research and development expenses/sales	0.564 (<0.001)	0.638 (<0.001)	0.597 (<0.001)	1.040 (<0.001)	3.107 (<0.001)	1.098 (<0.001)
Capital expenditures/book assets	–1.839 (<0.001)	–2.523 (<0.001)	–1.941 (<0.001)	–1.280 (0.061)	–1.590 (0.072)	–1.286 (0.031)
Dividend paying dummy	–0.125 (0.091)	–0.372 (<0.001)	–0.167 (0.006)	–0.319 (<0.001)	–0.203 (0.049)	–0.319 (<0.001)
Operating income/book assets	–0.061 (0.639)	0.579 (0.021)	0.026 (0.832)	–0.618 (0.130)	–1.182 (0.025)	–0.693 (0.059)
Total debt/book assets	–1.253 (<0.001)	–0.305 (0.654)	–1.107 (<0.001)	–0.700 (<0.001)	–1.810 (<0.001)	–0.695 (<0.001)
Industry cash flow risk	–0.158 (0.621)	0.517 (0.278)	–0.002 (0.993)	0.540 (0.186)	0.084 (0.801)	0.393 (0.227)
Acquisition expense/book assets	–1.663 (<0.001)	–3.349 (<0.001)	–2.008 (<0.001)	–2.586 (<0.001)	–3.968 (<0.001)	–2.684 (<0.001)
Commercial and industrial loan rate spread	–0.044 (0.796)	0.152 (0.570)	–0.034 (0.808)	–0.519 (0.018)	0.074 (0.836)	–0.384 (0.042)
Net debt issuance/book assets	2.271 (<0.001)	3.217 (<0.001)	2.523 (<0.001)	1.584 (<0.001)	0.802 (0.089)	1.589 (<0.001)
Firm had its IPO during the prior five years dummy	–0.110 (0.056)	0.023 (0.750)	–0.092 (0.064)	0.129 (0.729)	–0.126 (0.271)	0.011 (0.886)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N	13,207	3,895	17,102	4,979	1,421	6,400

Table VIII, Panel B: Financial constraints

Table IX
Debt Maturity and the Propensity to Save Cash Out of Current Cash Flows

Data are for Compustat industrial firms over the 1980 to 2008 period. The table reports OLS regressions of the change in cash holdings on current cash flow, a dummy variable for whether the fitted value for the fraction of long-term debt due in the next three years is in the top sample quintile over a particular year, an interaction term between this dummy variable and current cash flow, and control variables. The dependent variable is the change in cash and short-term investments scaled by book assets. Cash flow is earnings before extraordinary items and depreciation minus dividends scaled by book assets. The fitted value used in Model 2 is from a regression model explaining long-term debt due in the next three years/total long-term debt, where the independent variables are total debt/book assets, the natural logarithm of real book assets, market-to-book assets, the difference between the yield on a government 10-year and six-month bond, future year abnormal earnings, weighted average maturity of a firm's assets, industry cash flow risk, net debt issuance/book assets, whether a firm had its IPO during the prior five years, Fama-French (1997) 48 industry dummy variables, and year dummy variables. Cash holdings are not included in the change in net working capital variable. Significance levels for whether coefficient estimates are different from zero are in parentheses. The standard errors of the coefficients are adjusted for clustering of observations at the firm level.

Model	1	2
Intercept	0.043 (<0.001)	0.020 (<0.001)
Cash flow	0.040 (<0.001)	0.034 (<0.001)
Fitted value of long-term debt due fraction is in the top sample quintile		0.026 (<0.001)
Cash flow × fitted value of long-term debt due fraction is in the top sample quintile		0.017 (<0.001)
Market-to-book assets	0.010 (<0.001)	0.009 (<0.001)
Natural logarithm of real book value of assets	–0.002 (<0.001)	0.001 (<0.001)
Capital expenditures/book assets	–0.241 (<0.001)	–0.236 (<0.001)
Acquisition expense/book assets	–0.332 (<0.001)	–0.333 (<0.001)
Change in net working capital/book assets relative to prior year	–0.051 (<0.001)	–0.055 (<0.001)
Net debt issuance/book assets	0.045 (<0.001)	0.064 (<0.001)
Prior year cash holdings/book assets	–0.272 (<0.001)	–0.276 (<0.001)
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
F ² -adjusted	0.245	0.252
N	79,077	79,077

Table IX: Cash savings out of cash flow

6.4 Tables IX, X, and XI: cash savings, investment, and valuation

Read:

- Table IX shows that firms with high refinancing risk save more cash out of current cash flows.
- Table X shows that cash holdings are more important for investment among high-refinancing-risk firms, especially in tight credit markets.
- Table XI shows that the marginal value of cash is higher when firms face high refinancing risk.
- Together, these tests show that cash is not merely passively accumulated; it has real and valuation consequences.

Economic message: Refinancing risk turns cash into a valuable buffer. Cash allows firms to invest when debt rollover risk is high and is valued more by the market in those circumstances.

Table X
The Effect of Debt Maturity on the Importance of Cash Holdings for Investment

The table reports OLS regressions of investment on a dummy variable for whether the fraction of a firm's long-term debt due in the next three years is in the top sample quintile during a particular year, lagged cash holdings, the interaction of the two prior variables, net debt issuance/book assets, the natural logarithm of the real market value of assets, market-to-book assets, and preinvestment earnings/book assets. The sample is made up of firms included in our analysis of the market valuation of corporate cash holdings and consists of 56,252 firm-year observations over the 1980 to 2008 period with required data for the regressions. Investment is defined as the sum of capital expenditures, acquisition expenses, research and development expenses, and advertising expenses scaled by book assets. Preinvestment earnings/book assets is defined as earnings before interest, taxes, depreciation, and amortization plus research and development and advertising expenses scaled by book assets. The second model is the same as the first, except that it is run using data for only the five years during which the spread of commercial and industrial loan rates over the federal funds rate is in the highest quintile for the 29 years over the 1980 to 2008 sample period. Significance levels for whether coefficient estimates are different from zero are in parentheses. The standard errors of the coefficients are adjusted for clustering of observations at the firm level.

Model	Full Sample 1	Tight Credit Market Conditions 2
Constant	0.105 (<0.001)	0.144 (<0.001)
Fifth quintile of long-term debt due in next three years	-0.005 (0.004)	-0.016 (0.005)
Cash holdings _{<i>t-1</i>}	0.127 (<0.001)	0.121 (<0.001)
Fifth quintile of long-term debt due in next three years $\times$ Cash holdings _{<i>t-1</i>}	0.020 (0.005)	0.074 (<0.001)
Net debt issuance/book assets	0.214 (<0.001)	0.159 (<0.001)
Natural logarithm of real market value of assets	-0.001 (0.210)	-0.003 (0.162)
Market-to-book-assets	0.002 (<0.001)	-0.048 (0.002)
Preinvestment earnings/book assets	0.063 (<0.001)	0.144 (<0.001)
Year fixed effects	Yes	No
Firm fixed effects	Yes	Yes
R^2 -adjusted	0.167	0.177
N	56,252	10,153

Table X: Cash and investment

Table XI
The Effect of Debt Maturity on the Market Valuation of Cash Holdings

The table reports OLS regressions of changes in firm value on changes in cash holdings, a dummy variable for whether a firm's long-term debt has a short maturity, the interaction of the prior two variables, and control variables. The sample consists of 58,433 firm-year observations over the 1980 to 2008 period with required data for the regressions. The dependent and independent variables are defined as in Faulkender and Wang (2006). A delta (Δ) indicates that the variable is calculated as the change from year $t-1$ to t . The first model is the basic model from Faulkender and Wang (2006). In the second model, we include a dummy variable for whether the fraction of a firm's long-term debt due in the next three years is in the top sample quintile for that year, and also include the interaction of this dummy variable with Δ Cash holdings. The third model is the same as the second model, except that it is run using data for only the five years during which the spread of commercial and industrial loan rates over the federal funds rate is in the highest quintile for the 29 years over the 1980 to 2008 sample period. Significance levels for whether coefficient estimates are different from zero are in parentheses. The standard errors of the coefficients are adjusted for clustering of observations at the firm level.

Model	Full Sample		Tight Credit Market Conditions
	1	2	3
Constant	0.040 (<0.001)	0.058 (<0.001)	0.044 (<0.001)
Δ Cash holdings	1.201 (<0.001)	1.128 (<0.001)	1.264 (<0.001)
Fifth quintile of long-term debt due in next three years	-0.061 (<0.001)	-0.061 (<0.001)	-0.046 (<0.001)
Fifth quintile of long-term debt due in next three years $\times$ Δ Cash holdings	0.247 (<0.001)	0.247 (<0.001)	0.345 (<0.001)
Δ Earnings	0.664 (<0.001)	0.659 (<0.001)	0.672 (<0.001)
Δ Net assets	0.036 (<0.001)	0.036 (<0.001)	0.026 (<0.001)
Δ Research and development expenses	0.546 (<0.001)	0.506 (<0.001)	0.430 (<0.001)
Δ Interest expense	-1.557 (<0.001)	-1.511 (<0.001)	-1.352 (<0.001)
Δ Dividends	0.152 (0.689)	0.295 (0.004)	3.005 (<0.001)
Cash holdings _{<i>t-1</i>}	0.444 (<0.001)	0.444 (<0.001)	0.385 (<0.001)
Leverage	-0.430 (<0.001)	-0.457 (<0.001)	-0.324 (<0.001)
Net financing	0.238 (<0.001)	0.233 (<0.001)	0.110 (<0.001)
Cash holdings _{<i>t-1</i>} $\times$ Δ Cash holdings	-0.902 (<0.001)	-0.918 (<0.001)	-1.211 (<0.001)
Leverage $\times$ Δ Cash holdings	-0.730 (<0.001)	-0.616 (<0.001)	-0.526 (<0.001)
R^2 -adjusted	0.092	0.100	0.105

Table XI: Market valuation of cash

7 Short summary

- Firms face refinancing risk when a large portion of long-term debt matures soon.
- This risk can force costly refinancing, asset sales, liquidation, or underinvestment.
- The paper measures refinancing risk using the fraction of long-term debt due within three years.
- From 1980 to 2008, long-term debt maturities shortened while corporate cash holdings increased.
- The paper treats cash and debt maturity as jointly determined and estimates a simultaneous system.
- Shorter maturity debt significantly increases cash holdings.
- Shortening debt maturity explains roughly one-third of the secular increase in corporate cash reserves.
- The cash–debt maturity relation is stronger when credit markets are weak and when firms have more debt.
- Credit lines do not fully substitute for cash, especially for constrained firms.
- Firms with high refinancing risk save more cash from cash flow.
- Cash helps mitigate underinvestment among high-refinancing-risk firms.
- The market values cash more when refinancing risk is high.

8 Conclusion

8.1 Core conclusion

Harford, Klasa, and Maxwell show that refinancing risk is a central determinant of corporate cash holdings. Firms with more long-term debt maturing soon hold more cash, save more cash out of cash flow, and use cash to avoid underinvestment. The shortening of debt maturities over time helps explain the rise in corporate cash balances.

8.2 Economic intuition

Shorter debt maturity creates frequent rollover needs. If refinancing conditions are poor, firms may be forced to cut investment, sell assets, or liquidate inefficiently. Cash reduces the firm's dependence on external debt markets exactly when those markets may be least reliable. Because shorter maturity debt also subjects managers to more frequent monitoring, the agency cost of holding cash may be lower for these firms.

8.3 Takeaway

The paper's key lesson is that liquidity policy and debt maturity policy are complements. Firms do not simply choose cash holdings in isolation. They choose cash in response to refinancing risk created by their liability structure. Understanding corporate cash therefore requires understanding when debt comes due.